Ding Gao

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EDUCATION

Donghua University (211 Project University) | Shanghai, China

Master of Engineering in Intelligent Manufacturing | Sep 2023 – Present (Expected Graduation: June 2026)

- **GPA:** 3.93/4.0;
- **Awards:** First-Class and Second-Class University Scholarships (2023-2024, one each)
- **Competitions:** 2nd Place Excellence Award, 12th NVIDIA Sky Hackathon (Embodied AI Robot based on LLM Multi-Agent Systems)

Nanjing Institute of Technology | Nanjing, China

Bachelor of Engineering in Mechanical Engineering | Sep 2019 – June 2023

- **GPA:** 3.77/4.0 (Last Two Years); **Major Rank:** 4/99;
- **Awards:** University First-Class and Second-Class Scholarships (5 times, 2020-2023); University Merit Student (2022, 2023); Outstanding Graduate Award.
- **Competitions:** National First Prize, National College Students Mathematics Competition (2023); Provincial First Prize, Provincial College Students Mathematics Competition (2022); Provincial Third Prize, Provincial College Students Mathematics Competition (2020); National First Prize, 6th National College Students Crane Creativity Competition.
- **Thesis:** "Three-Degree-of-Freedom Ball Screw Defect Detection Platform"

RESEARCH EXPERIENCE

Research Assistant – Digital Twin & LLM-based Multi-Agent Systems | Donghua University, Shanghai, China
Advisor: Prof. Jinsong Bao | Sep 2023 – Present

- Designed and developed a digital twin system architecture integrating frontend (React, Next.js 14, ThreeJS/R3F), backend (Supabase), and multi-agent coordination (LangChain, LangGraph).
- Established real-time synchronization between physical production line units and their digital models using Socket/WebSocket protocols and ThreeJS for dynamic state updates.
- Proposed and implemented a three-model framework (Knowledge, Behavior, Geometry) for digital twins, utilizing LLM-driven multi-agent systems for semantic modeling of complex manufacturing.
- Developed LLM-driven agents featuring streaming inter-agent dialogue (Generator functions, ReadableStream), event-driven function calls for model manipulation (Event Bus), and integrated STT/TTS for enhanced voice command processing using voice-specialized LLMs.
- Implemented RAG-enhanced retrieval of production knowledge accessible via natural language interaction.

Research Assistant – Embedded AI Tool Recommendation Robot for Industrial Disassembly | Donghua University, Shanghai, China
Advisor: Dr. Jiahui Si | Apr 2025 – June 2025

- Developed a multi-modal perception-decision-action (VLA) system for disassembling retired EV batteries, enabling LLM-driven strategy generation and robotic arm control via voice commands.
- Constructed an LLM-based agent using LangGraph, integrated with Alibaba's Tongyi Qianwen voice model for real-time task planning and voice interaction.
- Utilized FastSAM for image segmentation and Visual Language Models (VLLMs) for scene understanding. Improved visual recognition accuracy for grasp planning using Strategy of Mind (SoM) prompting to reduce model hallucinations.
- Deployed GraspNet to generate collision-free grasp poses from point clouds and established MCP protocol communication for a UR robotic arm. The LLM agent utilized these poses via tool calling to control the arm for tool pickup/delivery tasks.

Undergraduate Thesis – 3-DOF Ball Screw Defect Detection Platform | Nanjing Institute of Technology, Nanjing, China
Advisor: A/Prof. Fei Hao | Jan 2023 – June 2023

- Performed axial/radial load and motor torque calculations for the ball screw based on technical parameters and selected cost-effective materials.
- Completed 2D engineering drawings and 3D modeling (CAD, UG) and conducted kinematic simulations in UG by configuring drivers and motion pairs.
- Programmed an 8051 microcontroller (C language) to generate PWM signals via internal interrupts for precise control of stepper motors enabling linear/arc motion.
- Designed circuit schematics in Proteus using selected components and validated system functionality through microcontroller behavior simulation.

National College Students Crane Creativity Competition (6th) | Nanjing Institute of Technology, Nanjing, China

Advisor: A/Prof. Fei Hao | Feb 2022 – Nov 2022

Description: Led the development of a crane robot for transporting various goods along a set route within a time limit.

- Designed an optimized mechanical structure, including a kinematics-calibrated mobile chassis and 3D printed sector mechanisms to compensate for STM32 interface limitations.
- Programmed an STM32 microcontroller (C language) to control DC motors, encoder motors, servos, and a robotic arm, coordinating modules via CAN bus for precise movement, grasping, and placement.
- Designed a path optimization algorithm to dynamically calculate the shortest route, improving pickup and placement efficiency.
- Upgraded the mechanism from single to triple DC electric actuator configuration, reducing total operation time from 4 min 32 sec to 3 min 15 sec while maintaining reliability.

PUBLICATIONS AND PRESENTATIONS

- Co-inventor, Utility Model Patent: "A Prefabricated Component System Based on Conductive Concrete" (Patent No. ZL 12601087).
- Co-inventor, Invention Patent: "An Angle-Adjustable Stage and Its Use Method" (Publication No. CN113702385A).
- Co-author, Lv J, Si J, Gao D, et al. Historical visual question answering with large language model for Augmented Reality-assisted Human–Robot Collaboration[J].Journal of Manufacturing Systems,2025,83546-556.DOI:10.1016/J.JMSY.2025.10.005.
- Co-author, Lv J, Si J, Li W, Gao D, et al. Graph-driven Single-Robot Multi-Cognitive Agent System architecture for human–robot collaborative disassembly[J].Robotics and Computer-Integrated Manufacturing,2026,99103207-103207.DOI:10.1016/J.RCIM.2025.103207.
- First-author, “Self-Evolving Digital Twins in Flexible Manufacturing: A Large Language Model-Powered Multi-Agent Approach” (Journal: Digital Engineering, Status: Major Revision).

PROFESSIONAL EXPERIENCE

TESLA GIGAFACTORY | SHANGHAI, CHINA

Software Development Intern | R&D Department | Oct 2025 – Jan 2026

- Data Analysis: Analyzed multi-channel user feedback for Model 3 and Model Y to quantify and prioritize critical issues and high-demand feature requests for product development cycles.
- Data Display: Developed a full-stack internal visualization platform using Angular and C# to enable teams to dynamically adjust project priorities based on real-time data insights.

ABB ROBOT SUPER FACTORY | SHANGHAI, CHINA

Software Development Intern – Robotics Application Software | R&D Department | Jul 2025 – Sep 2025

- Developed an LLM-driven task planner using Azure OpenAI and MCP protocol, capable of decomposing complex instructions into RAPID commands via RWS services to execute parallel tasks and handle execution failures.
- Enhanced robotic vision robustness for a Tangram solver by implementing dynamic HSV parameter adjustment scripts, solving recognition issues caused by variable lighting environments.
- Built a RAG-powered coding assistant within App Studio, enabling engineers to generate robot functionalities and retrieve technical insights through natural language interaction.

BILIBILI | SHANGHAI, CHINA

Front-End Development Intern | Jun 2025 – Jul 2025

- Developed H5/PC certification interfaces using Vue/Vite, managed state with VueX, and integrated RESTful APIs.
- Built AI-powered error monitoring: implemented Cron jobs (NestJS) to capture JS/Rejection errors, analyzed them using LLM agents, and stored results in MySQL.
- Created a RAG-based retrieval agent for automatic error diagnosis, outputting structured JSON solutions for front-end rendering.

CAERULEAN AI INC. | VANCOUVER, CANADA

Full-Stack Developer Intern (Remote) | Dec 2024 – Apr 2025

- Architected a multi-agent system within an interactive virtual world using LangChain, implementing memory-augmented agents with RAG-enhanced contextual interaction, integrated into the browser using Phaser.

PROFESSIONAL SKILLS

- **CAD/CAM:** Proficient: AutoCAD, SolidWorks, UG NX. Familiar: ANSYS, Pro/ENGINEER.
- **Programming Languages:** Proficient: Python, JavaScript/TypeScript. Familiar: C, C++, MATLAB.
- **Embedded Systems & Circuits:** ARM Microcontrollers (STM32), Keil MDK, STM32CubeMX, Proteus.
- **Web Development:** Full-Stack (React, Next.js) Development, Docker, Kubernetes Deployment.
- **Operating Systems:** Proficient in Windows, Familiar with Linux (Ubuntu), Mac OS.
- **Computer Science:** Data Structures & Algorithms, Computer Networks (TCP/IP, Socket programming).
- **Artificial Intelligence:** LLM(OpenAI API), Fine-Tuning, Deployment, RAG, MCP Protocol; Multi-Agent System Construction.
- **English : IELTS:** 7.0 Overall (Listening: 7.5, Reading: 8.0, Writing: 6.0, Speaking: 6.0).